

Aniruddha Kagalkar

Principal/Staff Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with 18+ years of experience architecting and implementing scalable data infrastructure, ETL pipelines, and cloud-native data platforms. Proven expertise in designing end-to-end data solutions using modern technologies including Kafka, Apache Airflow, Kubernetes, and cloud platforms. Successfully led cross-functional teams to deliver high-impact data engineering solutions that drive business intelligence and operational efficiency. Specialized in big data processing, real-time streaming, data warehouse optimization, and MLOps pipeline development.

Key Achievements

- Architected and implemented enterprise-scale ETL pipelines processing TB+ of data daily using Kafka, Apache Airflow, and Py-Faust, enabling real-time analytics for business-critical operations
 - Led data platform transformation by consolidating 25+ data repositories into unified data architecture, reducing data redundancy by 20% and improving processing efficiency by 35%
 - Designed cloud-native data infrastructure on AWS and Kubernetes, optimizing resource utilization and achieving \$120K annual cost savings
 - Built automated data quality frameworks and monitoring systems using Prometheus and custom Telegraf plugins, improving data reliability by 95%
 - Mentored 40+ engineers across data engineering, DevOps, and analytics teams, establishing best practices for data governance and pipeline development
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PROFESSIONAL EXPERIENCE

Senior Data Engineer | Atlassian India LLP | Pune, India

April 2024 - Present

Data Platform Leadership & ETL Optimization

- **What:** Led ETL and data reporting initiatives for Databricks-based analytics platform serving 1000+ internal stakeholders
- **How:** Implemented performance optimization strategies, data quality checks, and automated monitoring for critical data pipelines
- **Result:** Achieved 99.9% data pipeline uptime and reduced query response times by 40% for business intelligence dashboards

Data Architecture & Repository Consolidation

- **What:** Designed and executed monorepo transformation for 25+ Jira plugin data repositories
- **How:** Implemented unified data schemas, standardized ETL patterns, and automated data validation frameworks

- **Result:** Reduced code duplication by 20%, improved build efficiency by 35%, and decreased deployment time from hours to minutes

Cloud Cost Optimization & Infrastructure

- **What:** Developed AWS cost monitoring and optimization system for data infrastructure
- **How:** Implemented automated resource scaling, usage analytics, and cost allocation tracking using CloudWatch and custom metrics
- **Result:** Achieved \$120K annual savings while maintaining 99.95% system availability for data processing workloads

Test Automation & Data Quality

- **What:** Automated data pipeline testing and validation processes
- **How:** Converted 250+ test cases to TypeScript using LLM assistance and implemented Playwright-based testing framework
- **Result:** Reduced manual testing effort by 50% and improved data quality confidence by 70%

Staff Data Engineer | VMware India Pvt Ltd | Pune, India

April 2017 – April 2024

Enterprise Data Platform Architecture

- **What:** Architected comprehensive data platform supporting microservices ecosystem for 40+ engineering teams
- **How:** Designed layered architecture with data ingestion, processing, and serving layers using Kubernetes, Kafka, and MongoDB
- **Result:** Enabled real-time data processing for 50+ applications with sub-second latency and 99.9% availability

End-to-End ETL Pipeline Development

- **What:** Built enterprise-scale ETL pipeline collecting data from 100+ diverse sources
- **How:** Implemented streaming data architecture using Kafka, KSQL, Apache Airflow, and Py-Faust for real-time processing, with HBase for columnar storage and MapReduce jobs for large-scale batch processing
- **Result:** Processed 10TB+ daily data volume with 95% reduction in data processing latency and enabled real-time business analytics

Data Infrastructure as Code

- **What:** Developed cloud-native data infrastructure deployment and management system
- **How:** Created Kubernetes manifests, Helm charts, and Terraform modules for automated data platform provisioning
- **Result:** Reduced infrastructure deployment time from weeks to hours and achieved 100% infrastructure reproducibility

Database Design & Optimization

- **What:** Designed scalable database schemas for RDBMS, NoSQL systems, and HBase columnar storage supporting multi-terabyte data volumes
- **How:** Implemented optimized indexing strategies, HBase region splitting, partitioning schemes, and MapReduce-based data processing for high-performance data access
- **Result:** Improved query performance by 60% and reduced storage costs by 25% while supporting 10x data growth

Data Observability & Monitoring

- **What:** Implemented comprehensive data pipeline monitoring and observability system
- **How:** Deployed Prometheus-based monitoring with custom Telegraf plugins in Golang for data quality metrics
- **Result:** Achieved 99.9% data pipeline reliability and reduced mean time to detection (MTTD) for data issues by 80%

Principal Software Engineer | Symantec India Pvt Ltd | Pune, India

July 2015 – April 2017

Security Management Platform Development

- **What:** Led a team of 6+ engineers to develop a Java & PHP-based security management platform
- **How:** Built an automation framework with comprehensive testing and code quality improvements
- **Result:** Increased code coverage by 14% and improved platform reliability

Module Lead

Avaya India Pvt Ltd | Pune, India

April 2012 – July 2015

VoIP and Unified Communications Solutions

- **What:** Designed and implemented VoIP and unified communications solutions for real-time communication
- **How:** Developed an automation framework to enhance testing efficiency and reliability
- **Result:** Ensured seamless real-time communication and improved system performance

Senior Consultant

Capgemini India Pvt Ltd | Mumbai, India

January 2011 – April 2012

Enterprise System Integration

- **What:** Led system integration projects for seamless enterprise application communication
- **How:** Implemented integration frameworks and communication protocols between enterprise systems
- **Result:** Achieved seamless data flow and improved inter-application communication efficiency

Senior Software Engineer | Avaya India Pvt Ltd | Pune, India

March 2006 – December 2010

- **What:** Owned development of Java-based web applications and L3/L4 search engine for customer support
 - **How:** Leveraged historic PR and MR data to build search algorithms and enhance productivity tools
 - **Result:** Improved customer response time by 80% and enhanced productivity for Support, Documentation, and Development teams
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EDUCATION

Master of Science (Computer Applications)

Symbiosis Institute of Computer Studies and Research | 2004 – 2006

Bachelor of Computer Science

Willingdon College of Arts and Sciences, Sangli | 2001 – 2004

TECHNICAL SKILLS

Data Engineering & Processing

- Apache Kafka, Apache Airflow, Py-Faust, KSQL, Apache Spark
- HBase, MapReduce, Hadoop Ecosystem
- ETL/ELT Pipelines, Real-time Streaming, Batch Processing
- Data Warehousing, Data Lakes, Delta Lake

Programming & Development

- Python, Java, Golang, JavaScript, TypeScript, SQL
- Pandas, NumPy, PySpark, Kafka Streams
- REST APIs, Microservices Architecture

Cloud Platforms & Infrastructure

- AWS (S3, Redshift, EMR, Glue, Lambda, Kinesis)
- Kubernetes, Docker, Terraform, Helm
- Infrastructure as Code, GitOps

Databases & Storage

- **SQL:** PostgreSQL, MySQL, Oracle
- **NoSQL:** MongoDB, Cassandra, DynamoDB, HBase
- **Data Warehouses:** Redshift, Databricks, BigQuery
- **Big Data:** Hadoop, HDFS, MapReduce

DevOps & Monitoring

- CI/CD Pipelines (GitLab CI, Jenkins)
- Prometheus, Grafana, Telegraf, CloudWatch

- Data Quality Monitoring, Pipeline Orchestration
 - Testing & Automation
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Education

Master of Science (Computer Applications)

Symbiosis Institute of Computer Studies and Research (2004 – 2006)

Bachelor of Computer Science

Willingdon College of Arts and Sciences, Sangli (2001 – 2004)

Awards and Recognition

- **5 Kudos Awards (Atlassian)** – Technical leadership, infrastructure innovation, and cross-team collaboration
 - **2x Accelerate Our Best Award (VMware)** – Driving efficiency in cloud-native infrastructure applications
 - **Club Stellar Award (Avaya)** – Outstanding contributions to infrastructure and software engineering
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